

Linear Algebra 3

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1 Introduction

Let $\mathcal{A} : V \rightarrow V$ be a linear operator and $\{V_i\}_{i \in [k]}$ be $\mathcal{A}$ -invariant subspaces such that $V = \bigoplus_{i \in [k]} V_i$. Then $\ker \mathcal{A} = \bigoplus_{i \in [k]} \ker(\mathcal{A}|_{V_i})$.

Note that $\ker(\mathcal{A}|_{V_i}) < V_i$, so we can speak of their direct sum. Suppose $v \in \ker \mathcal{A}$, so $\mathcal{A}v = 0$. Since V is the sum of the V_i , we may write $v = v_1 + v_2 + \dots + v_k$, so $\mathcal{A}v = \sum \mathcal{A}v_i = \sum (\mathcal{A}|_{V_i})v_i$. Note that $(\mathcal{A}|_{V_i})v_i \in V_i$, so by uniqueness of the presentation of $\vec{0}$ in a direct sum, $(\mathcal{A}|_{V_i})v_i = 0$ for all $i \in [k]$. That is, v is the sum of vectors v_i such that $v_i \in \ker(\mathcal{A}|_{V_i})$.

Conversely, suppose $v_i \in \ker(\mathcal{A}|_{V_i})$. Then $\mathcal{A}v_i = (\mathcal{A}|_{V_i})v_i = 0$ and $\mathcal{A}v = \sum \mathcal{A}v_i = 0$, so $\bigoplus_{i \in [k]} \ker(\mathcal{A}|_{V_i}) < \ker \mathcal{A}$. Equality follows.

$\mathcal{N} = (\mathcal{A} - \lambda \mathcal{E})|_{V^\lambda(\mathcal{A})}$ is nilpotent over $V^\lambda(\mathcal{A})$ (it is implicit $V^\lambda(\mathcal{A})$ is $\mathcal{A} - \lambda \mathcal{E}$ -invariant). Since $V^\lambda(\mathcal{A})$ is finite-dimensional (we assume V is finite-dimensional), let $e_1, e_2, \dots, e_k$ be a basis. By definition of $V^\lambda(\mathcal{A})$, there exist integers $m_1, m_2, \dots, m_k$ such that $(\mathcal{A} - \lambda \mathcal{E})^{m_i} e_i = 0$. Since the set of m_i is finite, we may let $m = \max_{i \in [k]} m_i$, so $\mathcal{N}^m e_i = 0$ for all $i \in [k]$, so $\mathcal{N}^m v = 0$ for all $v \in V^\lambda(\mathcal{A})$. Hence, $\mathcal{N}$ is nilpotent, as desired.

Let $e \in V^\lambda(\mathcal{A})$ and $U = \langle e, \mathcal{N}e, \mathcal{N}^2e, \dots, \mathcal{N}^{m-1}e \rangle$ be a cyclic subspace of $V^\lambda(\mathcal{A})$ of order m . U is $\mathcal{N}$ -invariant. Consider the matrix of $\mathcal{N}$ in U with respect to the basis $(\mathcal{N}^{m-1}e, \mathcal{N}^{m-2}e, \dots, \mathcal{N}e, e)$. The matrix is of the form

$$\begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & & & & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}.$$

Consider the matrix of $\lambda \mathcal{E}$ over U . This is simply λI_m . The matrices add, so the matrix of $\mathcal{A} = \mathcal{N} + \lambda \mathcal{E}$ is $J(\lambda, m)$ (Jordan block of eigenvector λ , order m).

Suppose the characteristic polynomial splits over K . Then $V = \bigoplus_{i \in [k]} V^\lambda(\mathcal{A})$.

Then split $V^\lambda(\mathcal{A})$ into cyclic subspaces (so we have a direct sum of many small spaces). We can compute the matrices of these subspaces independently, and each is a Jordan block.

2 Minimal Polynomial

The minimal polynomial of the Jordan block $A = J(\lambda, m)$ is $m_A = (t - \lambda)^m$. Note that $N = A - \lambda I$ is nilpotent with height m , so t^m is an annihilating polynomial of N and none of its divisors are, so t^m is the minimal polynomial.

3 Tensor Product

Let V, W be vector spaces with V finite dimensional. Define $\otimes : V^* \times W \rightarrow \text{Hom}(V, W)$ by $(\alpha \otimes y)(x) = \alpha(x)y$ for $\alpha \in V^*$ and $y \in W$. Then $\text{Hom}(V, W) = V^* \otimes W$.

First we verify $\alpha \otimes y \in \text{Hom}(V, W)$. Suppose $x, x' \in V$. Then $(\alpha \otimes y)(x + x') = \alpha(x + x')y = (\alpha(x) + \alpha(x'))y = \alpha(x)y + \alpha(x')y = (\alpha \otimes y)(x) + (\alpha \otimes y)(x')$. For $a \in K$, $(\alpha \otimes y)(kx) = \alpha(kx)y = k\alpha(x)y = k(\alpha \otimes y)(x)$.

Second, we show that the $\epsilon_\alpha \otimes y_\beta$ are independent, for $\{\epsilon_\alpha\}$ the dual basis of $\{e_\alpha\}$ (this forces J to be finite). Suppose $\sum \lambda_{\alpha\beta} \epsilon_\alpha(x) y_\beta = 0$ for all $x \in V$. Then

$$\sum_{\beta} \left(\sum_{\alpha} \lambda_{\alpha\beta} \epsilon_\alpha(x) \right) y_\beta = 0 \rightarrow \sum_{\alpha} \lambda_{\alpha\beta} \epsilon_\alpha(x) = 0$$

for all $x \in V$, so we may take $x = e_\alpha$, so $\lambda_{\alpha\beta} = 0$ for all α, β .

Lastly, we show that the $\epsilon_\alpha \otimes y_\beta$ span $\text{Hom}(V, W)$. Let $\varphi \in \text{Hom}(V, W)$. For all $\alpha \in J$, pick finite K_α so that $\varphi(e_\alpha) = \sum_{\beta \in K_\alpha} \lambda_{\alpha\beta} f_\beta$ for some fixed $\{\lambda_{\alpha\beta}\}$ (with respect to α).

For $v \in V$, we may write as a finite sum (since J is finite) $v = \sum_{\alpha \in J} \epsilon_\alpha(v) e_\alpha$. Then

$$\begin{aligned} \varphi(v) &= \sum_{\alpha \in J} \epsilon_\alpha(v) \varphi(e_\alpha) = \sum_{\alpha \in J} \epsilon_\alpha(v) \sum_{\beta \in K_\alpha} \lambda_{\alpha\beta} f_\beta \\ &= \sum_{\alpha \in J} \sum_{\beta \in K_\alpha} \lambda_{\alpha\beta} \epsilon_\alpha(v) f_\beta. \end{aligned}$$

The set $\{\lambda_{\alpha\beta}\}$ is fixed with respect to v , so we write $\varphi \in \langle \epsilon_\alpha \otimes f_\beta \rangle$.

Let $\otimes : V \times W \rightarrow T$ be the tensor product and $\otimes' : V \times W \rightarrow T'$ with the extension property. Then $T \cong T'$.

Take $U = T$, so there exists unique $\psi : T' \rightarrow T$ such that $\otimes = \psi \otimes'$. Now by the extension property on T , there exists unique $\phi : T \rightarrow T'$ such that $\otimes' = \phi \otimes$. It follows that $\phi \psi \otimes' = \otimes'$ and $\psi \phi \otimes = \otimes$.

By uniqueness of extension, since $\phi \psi \otimes' = \text{id}_{T'} \otimes' = \otimes'$, we have $\phi \psi = \text{id}_{T'}$. Similarly, $\psi \phi = \text{id}_T$, so ϕ, ψ are inverses and $T \cong T'$.